

LOCALITY-EMERGENCE-01 — Frozen Record

Ticket status: NO PASS (open).

Governing sentence: Level 1 is the prize. It is not in hand. Levels 3–5 are not entitled to comment yet.

Record date: 2026-09-04. Workspace: /home/user/workspace/expzero/.

1. Three isolated ledgers

Ledger	Content	State
CONJECTURE	relationship → locality → geometry → perhaps eventually gravity	Unadjudicated
EXPERIMENT	Can coordinate-free relational dynamics produce robust operational locality?	Runnable in part; estimator thresholds not yet calibrated
EVIDENCE	See §5	NO PASS

A higher ledger may not rewrite a lower one. Levels 2–5 are LOCKED.

2. The gate

$$P = N_{\text{no-smuggling}} \wedge C_{\text{connected}} \wedge Q_{\text{probe}} \wedge T_{\text{persistent}} \wedge D_{\text{finite}} \wedge M_{\text{metric}} \wedge X_{\text{nondegenerate}} \wedge A_{\text{ablation}} \wedge \xi$$

Conjunction only. No weighted score, no majority vote, no "promising." Terminal outcomes restricted to PASS | FAIL | INSTRUMENT_INVALID.

INSTRUMENT_INVALID is claimed whenever a predicate fails on a **known positive control** at the same measurement resolution. Such a run is not evidence against the conjecture.

3. Control matrix (merged; expander is one adversary, not the whole requirement)

ID	Control	Must show
C0	Mean-field / all-to-all	Stays mean-field; no finite D
C1	Random k -uniform hypergraph / random-regular	Expander behavior; fails D_{finite}
C2	Frozen skeleton (constraints exist, do not update)	Does not supply operational persistence for free
C3	Bounded capacity, no clustering bias	Degree bound alone does not manufacture locality
P1–P3	Ring (1D), 2D lattice, 3D lattice — coordinates used only to build clauses, then discarded	Positive controls; define the calibration band

4. Estimator pipeline (as implemented)

Layer 0 substrate: $V = \{1..N\}$, $s_i \in \{0, 1\}$, 3-uniform not-all-equal clauses on $e_\alpha \subset V$. No $x^\mu, r, g_{\mu\nu}, G, 1/r^2$, no target dimension, no lattice template in the update generator.

Update generator $\mathcal{L}$: soft-constraint Metropolis. $E(s) = \#\{\text{monochromatic clauses}\}$; pick label i uniformly, propose $s_i \rightarrow 1 - s_i$, accept with $\min(1, e^{-\beta\Delta E})$. β is a rigidity parameter with no geometric content; Ω_{allowed} is the $\beta \rightarrow \infty$ ground-state manifold.

Primary observable, causal not correlational:

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estimated over R matched pairs under **common random numbers** — the label sequence and the acceptance uniforms are shared between the two counterfactual arms.

Consequence, verified in this program: the sham-intervention null is **identically 0.000**. There are no false positives; the detection floor is set by ensemble resolution $\approx 3/R$, not by statistical noise. This is why conservative thresholds of the form $\epsilon = 6\sqrt{0.5/R}$ were rejected as unjustified.

Neighborhood is reconstructed from M , never declared from the constraint skeleton:

Dimension is estimated **two independent ways that are allowed to disagree**: ball-growth $V_i(\tau)$ fitted as power-law-with-free-origin-offset against exponential, and spectral return probability of a random walk on the influence kernel.

5. Evidence entered this session

5.1 Instrument corrections (verified in-session)

- **4-clique counter was wrong by a factor of 2.** `_k4_full` double-counted; corrected to $K_4 = \frac{1}{6} \sum_{(u,v) \in E} e(N(u) \cap N(v))$ and **validated against networkx clique enumeration** (both give 204 on an $N = 120$ instance where the old counter gave 102).
- **Incremental K_4 /triangle bookkeeping went negative** because `_rem` toggled pair-edges in the same order as `_add` rather than the reverse. Fixed. Incremental vs full recomputation now agree exactly across three $(\gamma, \delta, \epsilon)$ settings: T 1881/1881, 1342/1342, 2133/2133; K4 51/51, 77/77, 65/65.

Any earlier structural number quoting K_4 is retracted.

5.2 Structural screen of the relational action

Action $S_{\text{rel}} = -\gamma T + \delta \sum_i k_i^2/N + \epsilon K_4$, where T = triangles in the projected pair graph. Proposals are **uniformly random triples** — the proposal distribution knows nothing about locality. $N = 1000$, density 3.0, 170 sweeps, 85-sweep anneal, identical seed and initial hypergraph throughout.

γ	δ	ϵ	accept	giant	comps	clustering	deg cv	deg max	T	K4	isolated
0.0	25	0.0	0.878	1.000	1	0.082	0.28	34	3867	17	0
1.0	25	0.0	0.016	0.863	136	0.529	1.80	171	50654	193400	134
1.0	25	0.5	0.078	0.775	216	0.350	1.35	93	23892	10052	212
1.0	25	1.0	0.226	0.979	22	0.239	0.96	82	11706	1448	21
1.0	25	2.0	0.424	1.000	1	0.123	0.45	48	5531	53	0

Reading. The K_4 term does exactly what it was introduced to do — it dissolves the clique/shard collapse and restores a single giant component — but it does so by destroying the same redundancy that was the candidate signature of locality. Clustering falls monotonically from 0.529 to 0.123 as connectivity is restored, and the $\epsilon = 2.0$ endpoint is statistically indistinguishable from the $\gamma = 0$ no-action baseline (clustering 0.123 vs 0.082, cv 0.45 vs 0.28).

There are two regimes and no third. **No configuration in this screen is simultaneously connected, degree-homogeneous, and redundancy-rich**, which is the prerequisite for $C \wedge X \wedge D$. No point earned the cost of the full gate battery.

This is consistent with the structural objection already on record: maximizing triangles at fixed degree is optimized by disjoint $(d+1)$ -cliques, i.e. shards. Triangle reward plus degree penalty **cannot** produce extended geometry, and K_4 suppression buys connectivity only by buying its way back to randomness.

5.3 Verdict for this mechanism family

$$S_{\text{rel}} = -\gamma T + \delta \sum k_i^2 / N + \epsilon K_4 \Rightarrow \text{FAIL}$$

Recorded as FAIL, not INSTRUMENT_INVALID: the diagnostics were validated against reference implementations and the positive controls do produce the qualitative distinction (ring/lattice power-law vs random exponential) at this resolution. The mechanism, not the instrument, is what failed here.

Note on a related implementation reviewed this session (RELATA, `src/lib/engine.ts`): its overlap-pressure rewiring draws candidate triples from a **2-hop neighborhood pool**, with fallback to global only when that pool is small. That biases the proposal distribution toward locality, so locality recovered under it is partly inserted rather than derived. It fails $N_{\text{no-smuggling}}$ as written. Restricting proposals to uniformly random triples is the minimum fix.

5.4 Carried forward, unverified

The following are reported from earlier turns of this program and were **not** re-derived in this session; they are recorded as claims, not receipts: $\beta \approx 0.7\text{--}1.0$ optimizes per-hop dynamic range ($\approx 3.6 \times$ decay per hop, so $R \gtrsim 3.6^h$); estimator E1 (first-arrival) is sensitivity-limited and pinned near $D_H \approx 0.8\text{--}0.9$ on a 3D lattice; estimator E2 (response decay) returns the correct qualitative verdicts but biased-high exponents. Re-verify before citing.

6. What is deliberately NOT frozen

Numerical thresholds — probe-overlap minimum, persistence σ , D_{eff} drift tolerance, ϵ — remain CALIBRATION-PENDING. They must be fixed against P1–P3 and C0–C3 at a common measurement resolution and frozen before the candidate mechanism is measured. Selecting attractive numbers now would replace geometric smuggling with statistical smuggling.

This is the one remaining blocker on runnability of the full gate.

7. Standing instruction

Do not optimize toward three dimensions, toward gravity, toward a lattice, or toward PASS. Run the machinery against: *does a robust finite-dimensional operational-locality phase exist?* If NO after a serious search, record NO. If YES only at a razor-thin tuned point, record the weaker result.